Class 12 Biology

Previous Year Question Papers

Annual Examination 2023

Answer all questions. Marks are shown in brackets.

1. State where the light reaction of photosynthesis takes place. [1 mark]

2. Explain the role of ATP and NADPH in the Calvin cycle. [3 marks]

3. Name the process by which water is split during photosynthesis and state what is released. [2 marks]

4. State the net gain of ATP from glycolysis. [1 mark]

5. Why is oxygen essential for the electron transport chain? [3 marks]

6. Differentiate between mitosis and meiosis. [4 marks]

7. What is the phenotypic ratio of a monohybrid cross between two heterozygous parents? [2 marks]

8. Describe the structure of DNA and state the base-pairing rule. [4 marks]

Annual Examination 2024

Answer all questions. Marks are shown in brackets.

1. Explain how light intensity, carbon dioxide concentration and temperature limit the rate of photosynthesis. [4 marks]

2. Name the enzyme that fixes carbon dioxide in the Calvin cycle. [1 mark]

3. Where does the Krebs cycle take place and what does it produce? [3 marks]

4. Compare aerobic and anaerobic respiration in muscle cells. [4 marks]

5. Explain the importance of crossing over in meiosis. [3 marks]

6. State Mendel's law of segregation and explain it with an example. [4 marks]

7. Explain why DNA replication is called semi-conservative. [3 marks]

8. What is the ten percent law in an ecosystem? [2 marks]

9. Describe the role of nitrogen-fixing bacteria in the nitrogen cycle. [2 marks]